

Permissions Cheat Sheet (chmod, chown)

Read, change and set who can read, write and run files and folders.

Reading the output of ls -l

```
$ ls -l deploy.sh
-rwxr-xr-x 1 mark devs 2048 Mar 4 09:12 deploy.sh
|  ← type: - = file, d = folder, l = link
rwx  ← user (owner): read, write, execute
r-x  ← group: read, execute (- = not allowed)
r-x  ← other (everyone else): read, execute
1 mark devs  ← link count (rarely matters), then owner and group
```

Octal numbers: r = 4, w = 2, x = 1 (add them: rwx = 7, r-x = 5, rw- = 6)

Number	Shows as	Typical use
755	rwxr-xr-x	Scripts, programs and normal folders: everyone can use, only owner changes
644	rw-r--r--	Normal files: everyone can read, only owner can write
600	rw-----	Private files: SSH keys, passwords, <code>~/.ssh/authorized_keys</code>
700	rwx-----	Private folders and scripts: <code>~/.ssh</code>
775	rwxrwxr-x	Shared team folders: owner and group can write

Changing permissions and owners

Command	Does	Command	Does
chmod 755 f	Sets exact permissions	chmod a+r f	Everyone (all) can read
chmod u+x f	Owner (user) can execute	chmod -R g+w dir	Adds group write, recursively
chmod g-w f	Group can no longer write	chown mark f	Changes the owner
chmod o= f	Others get no access at all	chown mark:devs f	Changes owner and group
chmod u=rw,go=r f	Sets each part exactly (= 644)	chgrp devs f	Changes only the group

Who: **u** user, **g** group, **o** other, **a** all. How: **+** add, **-** remove, **=** set exactly. Changing the owner needs **sudo**.

Files vs. folders

	On a file	On a folder
r	Read the contents	List the names inside (ls)
w	Change the contents	Create, delete and rename files inside (even ones you can't write)
x	Run it as a program	Enter it (cd) and reach files inside it

Special bits and umask

Bit	Shows as	Means
4755 setuid	rwsr-xr-x	Runs as the file's owner , not as you (e.g. <code>/usr/bin/passwd</code>)
2775 setgid	rw-rwsr-x	On a folder: new files inside get the folder's group
1777 sticky	rw-rwxrwt	On a folder: only a file's owner can delete it (e.g. <code>/tmp</code>)
umask 022		Bits removed from new files: files become 644, folders 755 (077 = 600/700)

A capital **S** or **T** means the special bit is set but the **x** under it is not.

Handy recipes

Command	Does
find dir -type d -exec chmod 755 {} +	Sets all folders to 755
find dir -type f -exec chmod 644 {} +	Sets all files to 644
chmod -R u+rwx,go+rX dir	Capital X : execute only on folders (and already-executable files)
sudo chown -R www-data:www-data /var/www	Gives a web folder to the web server user
chmod 700 ~/.ssh && chmod 600 ~/.ssh/*	Fixes SSH "bad permissions" errors
stat -c '%a %U:%G %n' file	Shows octal permissions, owner and group

Remember: never "fix" access with **chmod 777**. Give the right owner or group just the permission it needs.